

Locara Atlas

Enterprise Dataset Delivery Platform

MVP Architecture & Database Schema

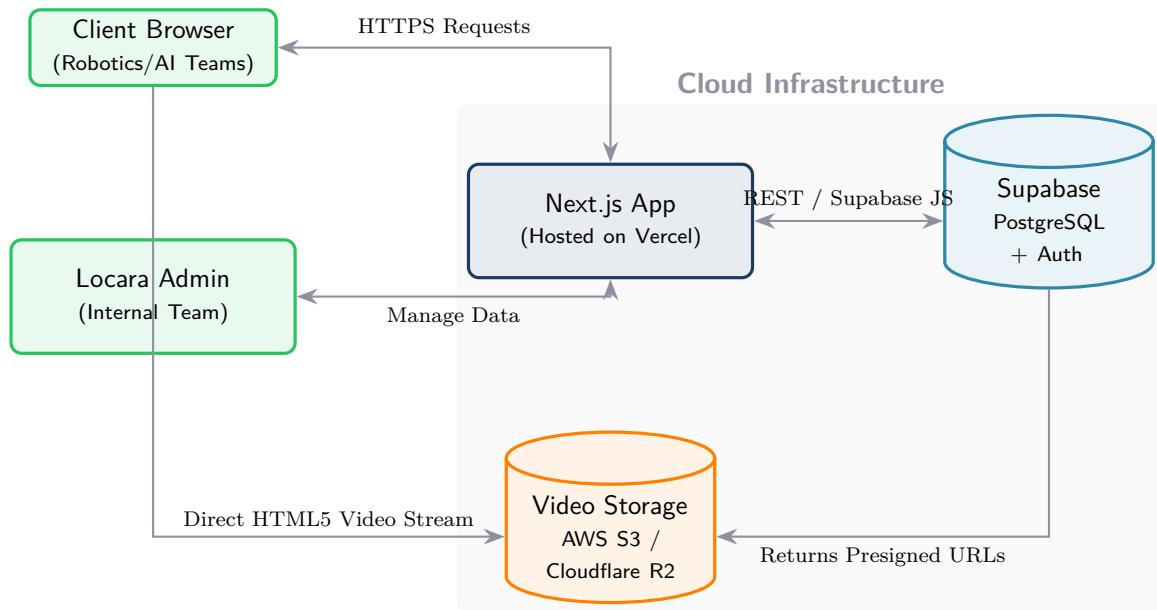
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Version: 1.0 (Interactive MVP Specification)
Date: June 16, 2026

Contents

1	System Architecture Overview	2
2	Database ER Diagram (Supabase PostgreSQL)	3
3	Row-Level Security (RLS) Structure	4
4	User Flow Diagram	4
5	Next.js UI Component Hierarchy	4

1 System Architecture Overview

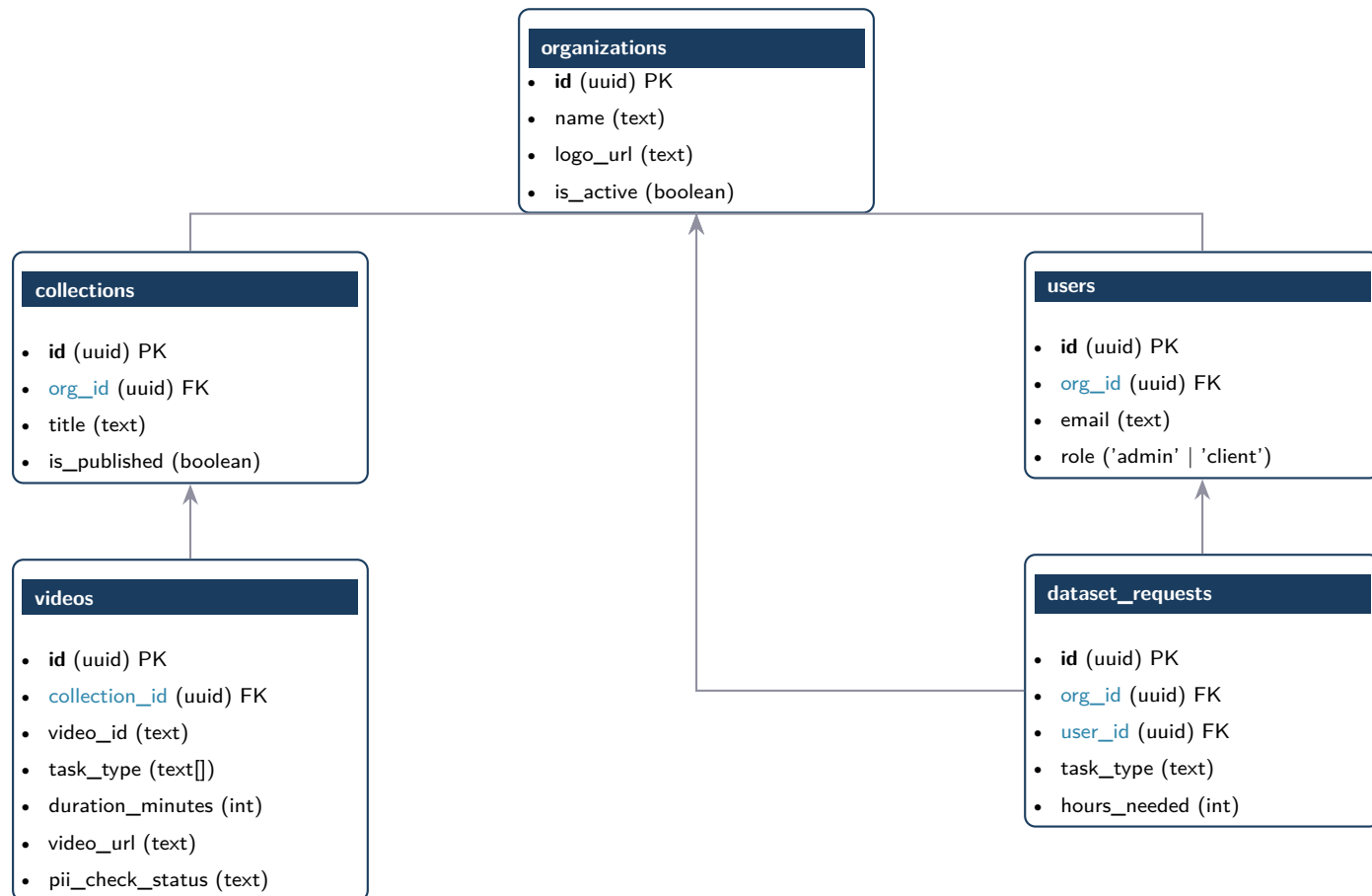
The Locara Atlas architecture is strictly decoupled to ensure the application remains lightning-fast even with 100,000+ videos.



Critical Architecture Decision: Video files (MP4s) are never stored inside the Next.js application, nor are they proxied through the server. The Supabase PostgreSQL database only stores the `video_url`. The client browser fetches the metadata from Supabase via Next.js, and then directly streams the massive video files from the S3/R2 storage bucket.

2 Database ER Diagram (Supabase PostgreSQL)

The backend strictly enforces tenant isolation via Row-Level Security (RLS). Below is the relational schema linking clients (organizations) to their assigned datasets.

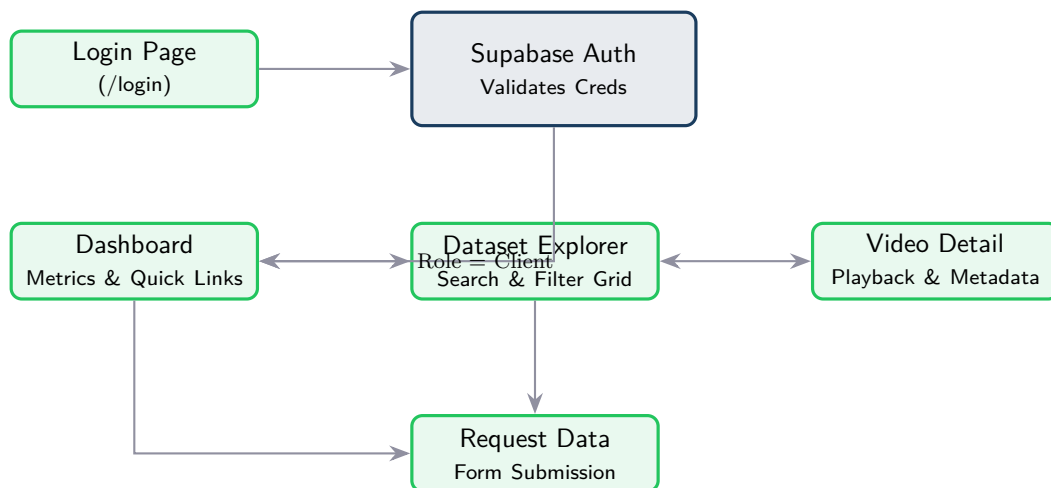


3 Row-Level Security (RLS) Structure

Supabase enforces RLS to guarantee data isolation. If a client user from *Measurement Labs* inspects the network payload or attempts a direct API call, the database itself will silently drop any rows belonging to *RIME*.

Table	Admin Role	Client Role
organizations	All Access	SELECT where <code>id = user's org_id</code>
users	All Access	SELECT where <code>org_id = user's org_id</code>
collections	All Access	SELECT where <code>org_id = user's org_id</code>
videos	All Access	SELECT where <code>collection_id ∈ client's collections</code>
dataset_requests	All Access	INSERT/SELECT where <code>org_id = user's org_id</code>

4 User Flow Diagram



5 Next.js UI Component Hierarchy

- **app/layout.tsx** (Root Layout + AuthProvider Context)
 - **app/(client)/layout.tsx** (Sidebar + Top Navbar)
 - **app/(client)/dashboard/page.tsx** (Recharts Analytics)
 - **app/(client)/collections/page.tsx** (Collection Grid)
 - **app/(client)/explorer/page.tsx** (Advanced Sidebar Filters + Video Grid)
 - **app/(client)/videos/[id]/page.tsx** (HTML5 Native Video Player)